fn sample_negative_binomial_rational

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This document proves that the implementation of `sample_negative_binomial_rational` in `mod.rs` satisfies its proof definition.

1 Hoare Triple

Precondition

`eta` $\in \mathbb{Q}_{>0}$ and `x` $\in \mathbb{Q}_{>0}$.

Pseudocode

```
def sample_negative_binomial_rational(eta, x) -> int:
    m = ceil(eta)
    eta_numer, eta_denom = Rational.into_numer_denom(eta)
    base_denom = eta_denom * m

    if eta_denom == 1:
        return sample_negative_binomial_integer(m, x)  #

    while True:
        k = sample_negative_binomial_integer(m, x)  #
        if sample_integer_envelope_accept(k, eta_numer, eta_denom, base_denom):  #
            return k
```

Postcondition

For any setting of the input parameters `eta` and `x` such that the given preconditions hold, `sample_negative_binomial_ration` either returns `Err(e)` due to a lack of system entropy, or `Ok(out)`, where `out` is distributed as a negative binomial random variable on $\mathbb{N}_0$ with shape `eta` and success probability $1 - e^{-x}$, parameterized as the number of failures before the `eta`-th success.

2 Proof

Assume the preconditions are met, and write $m := \lceil \texttt{eta} \rceil$. Define the target pmf

$$p_\eta(k) := (1 - e^{-x})^\eta e^{-xk} \prod_{i=0}^{k-1} \frac{\eta + i}{i + 1} \qquad \text{for } k \in \mathbb{N}_0.$$

This is the ordinary negative binomial pmf on $\mathbb{N}_0$ with shape η and success probability $1 - e^{-x}$.

Lemma 2.1. `sample_negative_binomial_rational` only returns `Err(e)` when there is a lack of system entropy.

Proof. If line 8 is executed, the only fallible call is to `sample_negative_binomial_integer`, which by its proof definition only returns `Err(e)` due to a lack of system entropy.

Otherwise, in each loop iteration line 11 invokes `sample_negative_binomial_integer`, and line 12 invokes `sample_integer_envelope_accept`. By their proof definitions, any returned error must be due to a lack of system entropy. There are no other fallible operations. □

Lemma 2.2. In the non-integer branch, each loop iteration draws a proposal $Y \sim NegativeBinomial(m, 1-e^{-x})$ on $\mathbb{N}_0$ and accepts it with probability $\prod_{i=0}^{Y-1} \frac{\eta+i}{m+i}$.

Proof. By the proof of `sample_negative_binomial_integer`, line 11 returns a proposal with pmf

$$\Pr[Y = k] = \binom{k+m-1}{k}(1-e^{-x})^m e^{-xk}.$$

By the proof of `sample_integer_envelope_accept`, conditional on $Y = k$, line 12 accepts with probability

$$\prod_{i=0}^{k-1} \frac{\eta+i}{m+i}.$$

□

Lemma 2.3. If the outcome of `sample_negative_binomial_rational` is `Ok(out)`, then `out` is distributed according to p_η.

Proof. If `eta` is an integer, then $m = \eta$ and the fast path on line 8 is taken. By the proof of `sample_negative_binomial_integ` the returned value already has pmf p_η.

Assume now that `eta` is not an integer. By Lemma 2.2, in each loop iteration and for every $k \in \mathbb{N}_0$,

$$\begin{aligned}
\Pr[Y = k \text{ and accepted}] &= \binom{k+m-1}{k}(1-e^{-x})^m e^{-xk} \prod_{i=0}^{k-1} \frac{\eta+i}{m+i} \\
&= (1-e^{-x})^m e^{-xk} \prod_{i=0}^{k-1} \frac{\eta+i}{i+1} \\
&= (1-e^{-x})^{m-\eta} p_\eta(k).
\end{aligned}$$

Summing over k shows that the total acceptance probability in each loop iteration is $(1-e^{-x})^{m-\eta}$, which is strictly positive because $x > 0$. Therefore the loop terminates almost surely. Conditioning on acceptance yields

$$\Pr[\texttt{out} = k] = \frac{(1-e^{-x})^{m-\eta} p_\eta(k)}{(1-e^{-x})^{m-\eta}} = p_\eta(k).$$

Thus the returned value has the claimed negative binomial distribution. □

Proof. 1 holds by Lemma 2.1 and Lemma 2.3. □